

Study Bot — AI Study Assistant Chatbot

Project overview :

A backend chatbot API that helps users with academic questions, remembers conversation history, and stores all chats in MongoDB Atlas. It uses a modern LLM (Groq/OpenAI compatible) for intelligent, context-aware answers.

- **Chat Endpoint:**
 - POST /chat
 - Accepts user_id and message, returns an AI-generated response.
- **Database Memory:**
 - Stores every message (user and assistant) in MongoDB with timestamps.
- **Context Awareness:**
 - Fetches the last 10 messages for each user to provide context to the LLM, making conversations feel natural and continuous.
- **Study Bot Behavior:**
 - System prompt ensures the bot only answers academic/educational questions and politely refuses others.
- **Error Handling:**
 - Handles missing environment variables, database errors, and LLM errors gracefully, returning clear HTTP responses.
- **Async FastAPI:**
 - High performance, supports many users at once, and is ready for frontend integration.

Explanation of memory implementation:

1. **Storing Messages (Saving Memory):**
 - Every time a user sends a message or receives a response from the assistant, both are saved in the MongoDB Atlas database.

- Each message is stored with:
 - [user_id](#) (to identify the user)
 - [role](#) ("user" or "assistant")
 - [message](#) (the text content)
 - timestamp (when the message was sent)

2. Retrieving Conversation History (Recalling Memory):

- When a new message comes in, the app fetches the last 10 messages for that user from the database.
- This history is sent to the LLM (AI model) along with the new message, so the AI can “see” the recent conversation and respond in context.

3. How It's Used in the Code:

- **save_message(user_id, role, message):**
Saves each message to the chats collection in MongoDB.
- **get_conversation_history(user_id, limit=10):**
Retrieves the last 10 messages for the user, sorted by time.
- **build_langchain_messages(history, current_message):**
Converts the history into the format expected by the LLM, including both user and assistant messages, plus the new user message.

4. Why This Matters:

- The bot can answer follow-up questions, remember what was discussed, and provide more natural, helpful responses.
- Without memory, every answer would be isolated and lack context.

GitHub Repository Link: <https://github.com/omkargouda1204/-Study-chat-.git>

Hosted API Link : <https://study-chat-mc87.onrender.com/docs>

Sample API Test Screenshots :

Responses

Curl

```
curl -X 'POST' \
  'https://study-chat-mc87.onrender.com/chat' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "user_id": "student",
    "message": "What is ai"
  }'
```

Request URL

https://study-chat-mc87.onrender.com/chat

Server response

Code

Details

200

Response body

```
{
  "response": "AI stands for Artificial Intelligence. It's a branch of computer science that focuses on creating intelligent machines that can perform tasks that would typically require human intelligence.\n\nImagine you're playing chess. A human player would analyze the board, think about the possible moves, and make a decision. An AI system can do the same thing, but instead of thinking like a human, it uses complex algorithms and data to make decisions.\n\nAI systems can:\n\n1. **Learn**: Like humans, AI systems can learn from data and improve their performance over time.\n2. **Reason**: AI systems can analyze information, identify patterns, and make decisions based on that analysis.\n3. **Perceive**: AI systems can take in information from sensors, images, or text and understand its meaning.\n4. **Act**: AI systems can take actions based on their analysis and decisions.\n\nThere are several types of AI, including:\n\n1. **Narrow or Weak AI**: Designed to perform a specific task, like a virtual assistant or a self-driving car.\n2. **General or Strong AI**: A hypothetical AI system that can perform any intellectual task that a human can.\n3. **Superintelligence**: A hypothetical AI system that is significantly more intelligent than the best human minds.\n\nSome examples of AI in our daily lives include:\n\n1. Virtual assistants like Siri, Alexa, or Google Assistant.\n2. Image recognition systems like facial recognition or object detection.\n3. Self-driving cars or drones.\n4. Personalized recommendations on social media or online shopping.\n5. Language translation tools.\n\nAs a student, you might be interested in AI because it's a rapidly growing field with many applications in science, technology, engineering, and mathematics (STEM)."
```

 Download

user_id

string

(path)

student

limit

integer

(query)

20

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'https://study-chat-mc87.onrender.com/history/student?limit=20' \
  -H 'accept: application/json'
```

Request URL

https://study-chat-mc87.onrender.com/history/student?limit=20

Server response

Code

Details

200

Response body

```
{
  "user_id": "student",
  "message_count": 2,
  "messages": [
    {
      "role": "user",
      "message": "What is ai",
      "timestamp": "2026-02-22T06:24:00.688000"
    },
    {
      "role": "assistant",
      "message": "AI stands for Artificial Intelligence. It's a branch of computer science that focuses on creating intelligent machines that can perform tasks that would typically require human intelligence.\n\nImagine you're playing chess. A human player would analyze the board, think about the possible moves, and make a decision. An AI system can do the same thing, but instead of thinking like a human, it uses complex algorithms and data to make decisions.\n\nAI systems can:\n\n1. **Learn**: Like humans, AI systems can learn from data and improve their performance over time.\n2. **Reason**: AI systems can analyze information, identify patterns, and make decisions based on that analysis.\n3. **Perceive**: AI systems can take in information from sensors, images, or text and understand its meaning.\n4. **Act**: AI systems can take actions based on their analysis and decisions.\n\nThere are several types of AI, including:\n\n1. **Narrow or Weak AI**: Designed to perform a specific task, like a virtual assistant or a self-driving car.\n2. **General or Strong AI**: A hypothetical AI system that can perform any intellectual task that a human can.\n3. **Superintelligence**: A hypothetical AI system that is significantly more intelligent than the best human minds.\n\nSome examples of AI in our daily lives include:\n\n1. Virtual assistants like Siri, Alexa, or Google Assistant.\n2. Image recognition systems like facial recognition or object detection.\n3. Self-driving cars or drones.\n4. Personalized recommendations on social media or online shopping.\n5. Language translation tools.\n\nAs a student, you might be interested in AI because it's a rapidly growing field with many applications in science, technology, engineering, and mathematics (STEM)."
```

Parameters

Cancel

Name	Description
user_id * required	
string	student12
(path)	

Execute

Clear

Responses

Curl

curl -X 'DELETE' \

'https://study-chat-mc87.onrender.com/history/student12' \

-H 'accept: application/json'

Request URL

https://study-chat-mc87.onrender.com/history/student12

Server response

Code

Details

200

Response body

{

"status": "success",

"message": "Cleared 2 messages for user student12"

}

Download

Response headers

access-control-allow-credentials: true

access-control-allow-origin: *

alt-svc: h3=":443"; ma=86400

cf-cache-status: DYNAMIC

cf-ray: 9d1c0d4ec5b9371-MAA

content-encoding: br

content-length: 61

default

^

GET

/ Root

^

Health check endpoint

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

curl -X 'GET' \

'https://study-chat-mc87.onrender.com/' \

-H 'accept: application/json'

Request URL

https://study-chat-mc87.onrender.com/

Server response

Code

Details

200

Response body

{

"status": "online",

"service": "AI Study Bot",

"version": "1.0.0",

"message": "Send POST requests to /chat to interact with the bot"

}

Download

Response headers

alt-svc: h3=":443"; ma=86400